

Accelerator ML Living Review

Summary Statistics

per_year: 12
per_category: 16
per_venue/journal: 218
per_keyword: 17
monthly_trends: 82

Papers

Machine learning as a service system for particle accelerator and its application in CSNS

Mei, Hao, Zhang, Yuliang, Peng, Na, Cheng, Sinong, He, Yongcheng, Xue, Kangjia, Wang, Lin, Li, Mingtao, Wu, Xuan, Zhu, Peng (2025)

Radiation Detection Technology and Methods

A generative adversarial network to improve integrated mode proton imaging resolution using paired proton-carbon data.

Simard M, Fullarton R, Volz L, Schuy C, Chung S, Baker C, Graeff C, Fekete CC. (2025)

Med Phys

A Time-Series Approach for Machine Learning-Based Patient-Specific Quality Assurance of Radiosurgery Plans

Buzzi S, Mancosu P, Bresolin A, Gallo P, La Fauci F, Lobefalo F, Paganini L, Pelizzoli M, Reggiori G, Franzese C, Tomatis S, Scorsetti M, Lenardi C, Lambri N. (2025)

Bioengineering (Basel)

A Supervised Machine Learning Framework for Multipactor Breakdown Prediction in High-Power Radio Frequency Devices and Accelerator Components: A Case Study in Planar Geometry

Asif Iqbal, John Verboncoeur, Peng Zhang (2025)

arXiv

Geoff: The Generic Optimization Framework & Frontend for Particle Accelerator Controls

Penelope Madysa, Sabrina Appel, Verena Kain, Michael Schenk (2025)

arXiv

Towards Agentic AI on Particle Accelerators

Antonin Sulc, Thorsten Hellert, Raimund Kammering, Hayden Hoschouer, Jason St. John (2025)

arXiv

Acceleration of Multi-Scale LTS Magnet Simulations with Neural Network Surrogate Models

Louis Denis, Julien Dular, Vincent Nuttens, Mariusz Wozniak, Benoît Vanderheyden, Christophe Geuzaine (2025)

arXiv

Application Of Large Language Models For The Extraction Of Information From Particle Accelerator Technical Documentation

Qing Dai, Rasmus Ischebeck, Maruisz Sapinski, Adam Grycner (2025)

arXiv

Predictive Hydrodynamic Simulations for Laser Direct-drive Implosion Experiments via Artificial Intelligence

Zixu Wang, Yuhan Wang, Junfei Ma, Fuyuan Wu, Junchi Yan, Xiaohui Yuan, Zhe Zhang, Jie Zhang (2025)

arXiv

Towards Foundation Models for Experimental Readout Systems Combining Discrete and Continuous Data

James Giroux, Cristiano Fanelli (2025)

arXiv

Towards generalizable deep ptychography neural networks

Albert Vong, Steven Henke, Oliver Hoidn, Hanna Ruth, Junjing Deng, Alexander Hexemer, Apurva Mehta, Arianna Gleason, Levi Hancock, Nicholas Schwarz (2025)

arXiv

FusionMAE: large-scale pretrained model to optimize and simplify diagnostic and control of fusion plasma

Zongyu Yang, Zhenghao Yang, Wenjing Tian, Jiyuan Li, Xiang Sun, Guohui Zheng, Songfen Liu, Niannian Wu, Rongpeng Li, Zhaohe Xu, Bo Li, Zhongbing Shi, Zhe Gao, Wei Chen, Xiaoquan Ji, Min Xu, Wulyu Zhong (2025)

arXiv

A Surrogate model for High Temperature Superconducting Magnets to Predict Current Distribution with Neural Network

Mianjun Xiao, Peng Song, Yulong Liu, Cedric Korte, Ziyang Xu, Jiale Gao, Jiaqi Lu, Haoyang Nie, Qiantong Deng, Timing Qu (2025)

arXiv

Machine Learning approach to classifying quench antenna signals

Plebani, Alberto, Barzi, Emanuela, Teyber, Reed (2025)

Thesis

Machine Learning for superconducting magnets application

Stabilini, Elisa (2025)

Thesis

Developing Machine Learning Models for Proton Computed Tomography and LHCb Particle Tracking

Ackernley, Thomas (2025)

Thesis

Batch spacing optimization by reinforcement learning

Remta, Matthias, Velotti, Francesco, Rezagholi, Sharwin (2025)

Phys.Rev.Accel.Beams

Road map for the tuning of hadronic interaction models with accelerator-based and astroparticle data

Albrecht, J., Becker Tjus, J., Behling, N., Blazek, J., Bleicher, M., Boelhauve, J., Cazon, L., Conceicao, R., Dembinski, H., Dietrich, L., Ebr, J., Ellbracht, J., Engel, R., Fedynitch, A., Fieg, M., Garzelli, M.V., Gaudu, C., Graziani, G., Gutjahr, P., Haungs, A., Huege, T., Hymon, K., Hünnefeld, Mirco, Kampert, K.-H., Kardum, L., Kolk, L., Korneeva, N., Kröninger, K., Maire, A., Menjo, H., Morejon, L., Ostapchenko, S., Paakkinen, P., Pierog, T., Plotko, P., Prosekin, A., Pyras, L., Pöschl, T., Rautenberg, Julian, Reininghaus, M., Rhode, W., Riehn, F., Roth, M., Sandrock, A., Sarcevic, I., Schmelling, M., Sigl, G., Sjöstrand, T., Soldin, D., Unger, M., Uthelm, M., Vícha, J., Werner, K., Windau, M.E., Zhukov, V. (2025)

arXiv

Self-supervised physics-informed generative networks for phase retrieval from a single X-ray hologram

Yang, Xiaogang, Hailu, Dawit, Kulvait, Vojtech, Jentschke, Thomas, Flenner, Silja, Greving, Imke, Campbell, Stuart I., Hagemann, Johannes, Schroer, Christian G., Wong, Tak Ming, Moosmann, Julian (2025)

Opt.Express

A Supervised Machine Learning Framework for Multipactor Breakdown Prediction in High-Power Radio Frequency Devices and Accelerator Components: A Case Study in Planar Geometry

Iqbal, Asif, Verboncoeur, John, Zhang, Peng (2025)
arXiv

Analysis note: measurement of thrust in e^+e^- collisions at $\sqrt{s} = 91$ GeV with archived ALEPH data

Badea, Anthony, Baty, Austin, Bossi, Hannah, Chen, Yu-Chen, Chen, Yi, Zhang, Jingyu, Innocenti, Gian Michele, Maggi, Marcello, McGinn, Chris, Peters, Michael, Sheng, Tzu-An, Mikuni, Vinicius, Avaylon, Matthew, Komiske, Patrick, Metodiev, Eric, Thaler, Jesse, Nachman, Benjamin, Lee, Yen-Jie (2025)
arXiv

Machine learning application for particle accelerator optimization-a review

Isti Dian Rachmawati, Nazrul Effendy, Taufik Taufik (2025)
IAES International Journal of Artificial Intelligence

Machine Learning for Particle Accelerators

Elena Fol, Auralee Edelen (2025)
Unknown Venue

FORT-GCN: A Fault-tolerant and Adaptive Accelerator Design for Efficient Graph Convolutional Network Inference

Ke Wang, Yingnan Zhao, Ahmed Louri (2025)
ACM Transactions on Embedded Computing Systems

(Invited) Analog Signal Processing Technologies for Power-Efficient Neural Network Inference and Training

Bert Jan Offrein (2025)
Meeting abstracts/Meeting abstracts (Electrochemical Society. CD-ROM)

Machine learning based parametrization of the resolution function for the first experimental area of the n_TOF facility at CERN

Petar Žugec, Marta Sabaté-Gilarte, Michael Bacak, Vasilis Vlachoudis, Adria Casanovas, Francisco García-Infantes (2025)
Nuclear Science and Techniques

-Diagnosis of Nasopalatine Duct and Nasopalatine Duct Cyst in CBCT Images: A Radiomics-Based Machine Learning Approach

H Duyan Yüksel, B Büyük, B Evlice (2025)
Dentomaxillofacial Radiology

Integration of Machine Learning-Based Plasma Acceleration Simulations into Geant4: A Case Study with the PALLAS Experiment

A. Sytov, K. Cassou, V. Kubyskyi, M. Lenivenko, A. Huber (2025)
arXiv

Optimisation of the Accelerator Control by Reinforcement Learning: A Simulation-Based Approach

Anwar Ibrahim, Denis Derkach, Alexey Petrenko, Fedor Ratnikov, Maxim Kaledin (2025)
arXiv

Physics-Informed Super-Resolution Diffusion for 6D Phase Space Diagnostics

Alexander Scheinker (2025)
arXiv

Optimizing Beam-Plasma Interactions Through Jitter Analysis Using Start-to-End Simulations

Robin Hwang (2024)
arXiv

Virtual Pulse Reconstruction Diagnostic for Single-Shot Measurement of Free Electron Laser Radiation Power

Till Korten, Vladimir Rybnikov, Peter Steinbach, Najmeh Mirian (2024)

arXiv

Harnessing Machine Learning for Single-Shot Measurement of Free Electron Laser Pulse Power

Till Korten, Vladimir Rybnikov, Mathias Vogt, Juliane Roensch-Schulenburg, Peter Steinbach, Najmeh Mirian (2024)

arXiv

Data-Driven Gradient Optimization for Field Emission Management in a Superconducting Radio-Frequency Linac

Steven Goldenberg, Kawser Ahammed, Adam Carpenter, Jiang Li, Riad Suleiman, Chris Tennant (2024)

arXiv

Data-Driven Discovery of Beam Centroid Dynamics

Liam A. Pocher, Irving Haber, Thomas M. Antonsen Jr., Patrick G. O'Shea (2024)

arXiv

Design and development of advanced Al-Ti-V alloys for beampipe applications in particle accelerators

Kamaljeet Singh, Kangkan Goswami, Raghunath Sahoo, Sumanta Samal (2025)

arXiv

Machine Learning for Reducing Noise in RF Control Signals at Industrial Accelerators

M. Henderson, J. P. Edelen, J. Einstein-Curtis, C. C. Hall, J. A. Diaz Cruz, A. L. Edelen (2024)

arXiv

Surrogate Models studies for laser-plasma accelerator electron source design through numerical optimisation

G. Kane, P. Drobniak, S. Kazamias, V. Kubytskyi, M. Lenivenko, B. Lucas, J. Serhal, K. Cassou, A. Beck, A. Specka, F. Massimo (2025)

arXiv

Beamline Steering Using Deep Learning Models

Dexter Allen, Isaac Kante, Dorian Bohler (2024)

arXiv

Beam-based Identification of Magnetic Field Errors in a Synchrotron using Deep Lie Map Networks

Conrad Caliri, Adrian Oeftiger, Oliver Boine-Frankenheim (2024)

arXiv

Linac_Gen: integrating machine learning and particle-in-cell methods for enhanced beam dynamics at Fermilab

Abhishek Pathak (2024)

arXiv

Automated Anomaly Detection on European XFEL Klystrons

Antonin Sulc, Annika Eichler, Tim Wilksen (2024)

arXiv

Accelerator beam phase space tomography using machine learning to account for variations in beamline components

Andrzej Wolski, Diego Botelho, David Dunning, Amelia E. Pollard (2024)

arXiv

Large Language Models for Human-Machine Collaborative Particle Accelerator Tuning through Natural Language

Jan Kaiser, Annika Eichler, Anne Lauscher (2024)

arXiv

Accelerating Cavity Fault Prediction Using Deep Learning at Jefferson Laboratory

Monibor Rahman, Adam Carpenter, Khan Iftekharuddin, Chris Tennant (2024)

arXiv

Anomaly Detection of Particle Orbit in Accelerator using LSTM Deep Learning Technology

Zhiyuan Chen, Wei Lu, Radhika Bhong, Yimin Hu, Brian Freeman, Adam Carpenter (2024)

arXiv

Machine-learning approach for operating electron beam at KEK S_e^-/e^+ injector Linac

Gaku Mitsuka, Shinnosuke Kato, Naoko Iida, Takuya Natsui, Masanori Satoh (2024)

arXiv

Cheetah: Bridging the Gap Between Machine Learning and Particle Accelerator Physics with High-Speed, Differentiable Simulations

Jan Kaiser, Chenran Xu, Annika Eichler, Andrea Santamaria Garcia (2024)

arXiv

Beyond PID Controllers: PPO with Neuralized PID Policy for Proton Beam Intensity Control in Mu2e

Chenwei Xu, Jerry Yao-Chieh Hu, Aakaash Narayanan, Mattson Thieme, Vladimir Nagaslaev, Mark Austin, Jeremy Arnold, Jose Berlioz, Pierrick Hanlet, Aisha Ibrahim, Dennis Nicklaus, Jovan Mitrevski, Jason Michael St. John, Gauri Pradhan, Andrea Saewert, Kiyomi Seiya, Brian Schupbach, Randy Thurman-Keup, Nhan Tran, Rui Shi, Seda Ogrenci, Alexis Maya-Isabelle Shuping, Kyle Hazelwood, Han Liu (2023)

arXiv

Robust Errant Beam Prognostics with Conditional Modeling for Particle Accelerators

Kishansingh Rajput, Malachi Schram, Willem Blokland, Yasir Alanazi, Pradeep Ramuhalli, Alexander Zhukov, Charles Peters, Ricardo Vilalta (2024)

arXiv

Machine Learning For Beamline Steering

Isaac Kante (2023)

arXiv

Variational Autoencoders for Noise Reduction in Industrial LLRF Systems

J. P. Edelen, M. J. Henderson, J. Einstein-Curtis, C. C. Hall, J. A. Diaz Cruz, A. L. Edelen (2023)

arXiv

Resilient VAE: Unsupervised Anomaly Detection at the SLAC Linac Coherent Light Source

Ryan Humble, William Collocho, Finn O'Shea, Daniel Ratner, Eric Darve (2023)

arXiv

Time-drift Aware RF Optimization with Machine Learning Techniques

R. Sharankova, M. Mwaniki, K. Seiya, M. Wesley (2023)

arXiv

Distance Preserving Machine Learning for Uncertainty Aware Accelerator Capacitance Predictions

Steven Goldenberg, Malachi Schram, Kishansingh Rajput, Thomas Britton, Chris Pappas, Dan Lu, Jared Walden, Majdi I. Radaideh, Sarah Cousineau, Sudarshan Harave (2023)

arXiv

Learning to Do or Learning While Doing: Reinforcement Learning and Bayesian Optimisation for Online Continuous Tuning

Jan Kaiser, Chenran Xu, Annika Eichler, Andrea Santamaria Garcia, Oliver Stein, Erik Bründermann, Willi Kuroпка, Hannes Dinter, Frank Mayet, Thomas Vinatier, Florian Burkart, Holger Schlarb (2023)

arXiv

From Compact Plasma Particle Sources to Advanced Accelerators with Modeling at Exascale

Axel Huebl, Remi Lehe, Edoardo Zoni, Olga Shapoval, Ryan T. Sandberg, Marco Garten, Arianna

Formenti, Revathi Jambunathan, Prabhat Kumar, Kevin Gott, Andrew Myers, Weiqun Zhang, Ann Almgren, Chad E. Mitchell, Ji Qiang, David Grote, Alexander Sinn, Severin Diederichs, Maxence Thevenet, Luca Fedeli, Thomas Clark, Neil Zaim, Henri Vincenti, Jean-Luc Vay (2023)
arXiv

Forecasting Particle Accelerator Interruptions Using Logistic LASSO Regression

Sichen Li, Jochem Snuerink, Fernando Perez-Cruz, Andreas Adelman (2023)

arXiv

Learning Electron Bunch Distribution along a FEL Beamline by Normalising Flows

Anna Willmann, Jurjen Couperus Cabada, Yen-Yu Chang, Richard Pausch, Amin Ghaith, Alexander Debus, Arie Irman, Michael Bussmann, Ulrich Schramm, Nico Hoffmann (2023)

arXiv

Physics-constrained 3D Convolutional Neural Networks for Electrodynamics

Alexander Scheinker, Reju Pokharel (2023)

arXiv

Identification of Magnetic Field Errors in Synchrotrons based on Deep Lie Map Networks

Conrad Caliri, Adrian Oeftiger, Oliver Boine-Frankenheim (2023)

arXiv

Data-driven Science and Machine Learning Methods in Laser-Plasma Physics

Andreas Döpp, Christoph Eberle, Sunny Howard, Faran Irshad, Jinpu Lin, Matthew Streeter (2023)

arXiv

Applications of Differentiable Physics Simulations in Particle Accelerator Modeling

Ryan Roussel, Auralee Edelen (2022)

arXiv

Prior-mean-assisted Bayesian optimization application on FRIB Front-End tuning

Kilean Hwang, Tomofumi Maruta, Alexander Plastun, Kei Fukushima, Tong Zhang, Qiang Zhao, Peter Ostroumov, Yue Hao (2022)

arXiv

Fault Prognosis in Particle Accelerator Power Electronics Using Ensemble Learning

Majdi I. Radaideh, Chris Pappas, Mark Wezensky, Pradeep Ramuhalli, Sarah Cousineau (2022)

arXiv

Machine learning-based analysis of experimental electron beams and gamma energy distributions

M. Yadav, M. Oruganti, S. Zhang, B. Naranjo, G. Andonian, Y. Zhuang, Ö. Apsimon, C. P. Welsch, J. B. Rosenzweig (2023)

arXiv

Review of Time Series Forecasting Methods and Their Applications to Particle Accelerators

Sichen Li, Andreas Adelman (2022)

arXiv

Automatic setup of 18 MeV electron beamline using machine learning

Francesco Maria Velotti, Brennan Goddard, Verena Kain, Rebecca Ramjiawan, Giovanni Zevi Della Porta, Simon Hirlander (2022)

arXiv

Diagnostics for Linac Optimization With Machine Learning

R. Sharankova, M. Mwaniki, K. Seiya, M. Wesley (2022)

arXiv

Transverse phase space tomography in the CLARA accelerator test facility using image compression and machine learning

Andrzej Wolski, Mark A. Johnson, Matthew King, Boris L. Militsyn, Peter H. Williams (2022)

arXiv

Using Kernel-Based Statistical Distance to Study the Dynamics of Charged Particle Beams in Particle-Based Simulation Codes

Chad E. Mitchell, Robert D. Ryne, Kilean Hwang (2022)

arXiv

Neural Network Solver for Coherent Synchrotron Radiation Wakefield Calculations in Accelerator-based Charged Particle Beams

Auralee Edelen, Christopher Mayes (2022)

arXiv

Adaptive Machine Learning for Time-Varying Systems: Towards 6D Phase Space Diagnostics of Short Intense Charged Particle Beams

Alexander Scheinker, Spencer Gessner (2022)

arXiv

Differentiable Preisach Modeling for Characterization and Optimization of Accelerator Systems with Hysteresis

R. Roussel, A. Edelen, D. Ratner, K. Dubey, J. P. Gonzalez-Aguilera, Y. K. Kim, N. Kuklev (2022)

arXiv

Mixed Diagnostics for Longitudinal Properties of Electron Bunches in a Free-Electron Laser

J. Zhu, N. M. Lockmann, M. K. Czwalińska, H. Schlarb (2022)

arXiv

A Neural Network Model of a Quasi-Periodic Elliptically Polarizing Undulator in Universal Mode

Ryan Sheppard, Cameron Baribeau, Tor Pedersen, Mark Boland, Drew Bertwistle (2022)

arXiv

Physics-informed neural network method for modelling beam-wall interactions

Kazuhiro Fujita (2022)

arXiv

Anomaly Detection in Particle Accelerators using Autoencoders

Jonathan P. Edelen, Nathan M. Cook (2021)

arXiv

Input Beam Matching and Beam Dynamics Design Optimization of the IsoDAR RFQ using Statistical and Machine Learning Techniques

Daniel Koser, Loyd Waites, Daniel Winklehner, Matthias Frey, Andreas Adelmann, Janet Conrad (2021)

arXiv

Neural Networks for ID Gap Orbit Distortion Compensation in PETRA III

Bianca Veglia, Ilya Agapov, Joachim Keil (2024)

arXiv

Time-Delayed Koopman Network-Based Model Predictive Control for the FRIB RFQ

Jinyu Wan, Shen Zhao, Wei Chang, Yue Hao (2024)

arXiv

Uncertainty Aware ML-based surrogate models for particle accelerators: A Study at the Fermilab Booster Accelerator Complex

Malachi Schram, Kishansingh Rajput, Karthik Somayaji Peng Li, Jason St. John, Himanshu Sharma (2022)

arXiv

Quantifying Uncertainty for Machine Learning Based Diagnostic

Owen Convery, Lewis Smith, Yarin Gal, Adi Hanuka (2021)

arXiv

Uncertainty Quantification for Virtual Diagnostic of Particle Accelerators

Owen Convery, Lewis Smith, Yarin Gal, Adi Hanuka (2021)

arXiv

Adaptive Latent Space Tuning for Non-Stationary Distributions

Alexander Scheinker, Frederick Cropp, Sergio Paiagua, Daniele Filippetto (2021)

arXiv

Adaptive deep learning for time-varying systems with hidden parameters: Predicting changing input beam distributions of compact particle accelerators

Alexander Scheinker, Frederick Cropp, Sergio Paiagua, Daniele Filippetto (2021)

arXiv

Using LSTM recurrent neural networks for monitoring the LHC superconducting magnets

Maciej Wielgosz, Andrzej Skoczko, Matej Mertik (2017)

arXiv

Coincident Learning for Beam-based RF Station Fault Identification Using Phase Information at the SLAC Linac Coherent Light Source

Jia Liang, William Colucho, Franz-Josef Decker, Ryan Humble, Ben Morris, Finn H. O'Shea, David A. Steele, Zhe Zhang, Eric Darve, Daniel Ratner (2025)

arXiv

Using Convolutional Neural Networks to Accelerate 3D Coherent Synchrotron Radiation Computations

Christopher Leon, Petr M. Anisimov, Nikolai Yampolsky, Alexander Scheinker (2025)

arXiv

Explainable physics-based constraints on reinforcement learning for accelerator controls

Jonathan Colen, Malachi Schram, Kishansingh Rajput, Armen Kasparian (2025)

arXiv

Accelerator system parameter estimation using variational autoencoded latent regression

Mahindra Rautela, Alan Williams, Alexander Scheinker (2024)

arXiv

Long Short-Term Memory Networks for Anomaly Detection in Magnet Power Supplies of Particle Accelerators

Ihar Lobach, Michael Borland (2024)

arXiv

Leveraging Prior Mean Models for Faster Bayesian Optimization of Particle Accelerators

Tobias Boltz, Jose L. Martinez, Connie Xu, Kathryn R. L. Baker, Zihan Zhu, Jenny Morgan, Ryan Roussel, Daniel Ratner, Brahim Mustapha, Auralee L. Edelen (2025)

arXiv

Optimizing Dynamic Aperture Studies with Active Learning

D. Di Croce, M. Giovannozzi, E. Krymova, T. Pieloni, S. Redaelli, M. Seidel, R. Tomás, F. F. Van der Veken (2024)

arXiv

Neural Network Prior Mean for Particle Accelerator Injector Tuning

Connie Xu, Ryan Roussel, Auralee Edelen (2022)

arXiv

Applications of object detection networks at high-power laser systems and experiments

Jinpu Lin, Florian Haberstroh, Stefan Karsch, Andreas Döpp (2022)

arXiv

A Neural Network approach to reconstructing SuperKEKB beam parameters from beamstrahlung

S. Di Carlo, G. Bonvicini, N. A. Althubiti, R. Ayad, E. De La Cruz-Burelo, I. Domínguez, B. O. El Bashir, H. Farhat, J. Flanagan, R. Gillard, S. Izaguirre Gamez, K. Kanazawa, K. Kumara, D. Liventsev, P. L. M. Podesta-Lerma, D. Ricalde-Herrmann, D. Rodriguez Perez, G. Tejeda-Muñoz,

M. Tobiyama I. Heredia de la Cruz (2022)
arXiv

Optimizing a Superconducting Radiofrequency Gun Using Deep Reinforcement Learning
David Meier, Luis Vera Ramirez, Jens Völker, Bernhard Sick, Jens Viefhaus, Gregor Hartmann (2022)
arXiv

Uncertainty aware anomaly detection to predict errant beam pulses in the SNS accelerator
Willem Blokland, Pradeep Ramuhalli, Charles Peters, Yigit Yucesan, Alexander Zhukov, Malachi Schram, Kishansingh Rajput, Torri Jeske (2021)
arXiv

Adaptive Machine Learning for Time-Varying Systems: Low Dimensional Latent Space Tuning
Alexander Scheinker (2021)
arXiv

Fast, efficient and flexible particle accelerator optimisation using densely connected and invertible neural networks
Renato Bellotti, Romana Boiger, Andreas Adelmann (2021)
arXiv

Invertible Surrogate Models: Joint surrogate modelling and reconstruction of Laser-Wakefield Acceleration by invertible neural networks
Friedrich Bethke, Richard Pausch, Patrick Stiller, Alexander Debus, Michael Bussmann, Nico Hoffmann (2021)
arXiv

Improving Surrogate Model Accuracy for the LCLS-II Injector Frontend Using Convolutional Neural Networks and Transfer Learning
Lipi Gupta, Auralee Edelen, Nicole Neveu, Aashwin Mishra, Christopher Mayes, Young-Kee Kim (2021)
arXiv

A Novel Approach for Classification and Forecasting of Time Series in Particle Accelerators
Sichen Li, Mélissa Zacharias, Jochem Snuverink, Jaime Coello de Portugal, Fernando Perez-Cruz, Davide Reggiani, Andreas Adelmann (2021)
arXiv

Real-time Artificial Intelligence for Accelerator Control: A Study at the Fermilab Booster
Jason St. John, Christian Herwig, Diana Kafkes, Jovan Mitrevski, William A. Pellico, Gabriel N. Perdue, Andres Quintero-Parra, Brian A. Schupbach, Kiyomi Seiya, Nhan Tran, Malachi Schram, Javier M. Duarte, Yunzhi Huang, Rachael Keller (2021)
arXiv

Surrogate Modeling of the CLIC Final-Focus System using Artificial Neural Networks
J. Ogren, C. Gohil, D. Schulte (2021)
arXiv

Physics-Based Deep Neural Networks for Beam Dynamics in Charged Particle Accelerators
Andrei Ivanov, Ilya Agapov (2020)
arXiv

Introduction to Machine Learning for Accelerator Physics
Daniel Ratner (2020)
arXiv

Machine learning for design optimization of storage ring nonlinear dynamics
Faya Wang, Minghao Song, Auralee Edelen, Xiaobiao Huang (2019)
arXiv

Studies in Applying Machine Learning to LLRF and Resonance Control in Superconducting RF Cavities

Jorge Alberto Diaz Cruz, Sandra Biedron, Manel Martinez-Ramon, Salvador Sosa, Reza Pirayesh (2019)
arXiv

The model of an anomaly detector for HiLumi LHC magnets based on Recurrent Neural Networks and adaptive quantization

Maciej Wielgosz, Matej Mertik, Andrzej Skoczko, Ernesto De Matteis (2017)
arXiv

Machine learning for analysis of plasma driven ion source

N. Joshi, O. Meusel, H. Podlech (2018)
arXiv

First Steps Toward Incorporating Image Based Diagnostics Into Particle Accelerator Control Systems Using Convolutional Neural Networks

A. L. Edelen, S. G. Biedron, S. V. Milton, J. P. Edelen (2016)
arXiv

Neural Networks for Modeling and Control of Particle Accelerators

A. L. Edelen, S. G. Biedron, B. E. Chase, D. Edstrom, S. V. Milton, P. Stabile (2016)
arXiv

Harnessing the Power of Gradient-Based Simulations for Multi-Objective Optimization in Particle Accelerators

Kishansingh Rajput, Malachi Schram, Auralee Edelen, Jonathan Colen, Armen Kasparian, Ryan Roussel, Adam Carpenter, He Zhang, Jay Benesch (2024)
arXiv

BOOSTR: A Dataset for Accelerator Control Systems

Diana Kafkes, Jason St. John (2021)
arXiv

Model-free and Bayesian Ensembling Model-based Deep Reinforcement Learning for Particle Accelerator Control Demonstrated on the FERMI FEL

Simon Hirlaender, Niky Bruchon (2022)
arXiv

Autonomous Control of a Particle Accelerator using Deep Reinforcement Learning

Xiaoying Pang, Sunil Thulasidasan, Larry Rybarcyk (2020)
arXiv

AI-Assisted Transport of Radioactive Ion Beams

Sergio Lopez-Caceres, Daniel Santiago-Gonzalez (2025)
arXiv

Bayesian Optimization Algorithms for Accelerator Physics

Ryan Roussel, Auralee L. Edelen, Tobias Boltz, Dylan Kennedy, Zhe Zhang, Fuhao Ji, Xiaobiao Huang, Daniel Ratner, Andrea Santamaria Garcia, Chenran Xu, Jan Kaiser, Angel Ferran Pousa, Annika Eichler, Jannis O. Lubsen, Natalie M. Isenberg, Yuan Gao, Nikita Kuklev, Jose Martinez, Brahim Mustapha, Verena Kain, Weijian Lin, Simone Maria Liuzzo, Jason St. John, Matthew J. V. Streeter, Remi Lehe, Willie Neiswanger (2024)
arXiv

SPIRAL2 Cryomodules Models: a Gateway to Process Control and Machine Learning

Adrien Vassal, Adnan Ghribi, François Millet, François Bonne, Patrick Bonnay, Pierre-Emmanuel Bernaudin (2021)
arXiv

Predicting Beam Transmission Using 2-Dimensional Phase Space Projections Of Hadron Accelerators

Anthony Tran, Yue Hao, Brahim Mustapha, Jose L. Martinex Marin (2022)

arXiv

Machine Learning for Orders of Magnitude Speedup in Multi-Objective Optimization of Particle Accelerator Systems

Auralee Edelen, Nicole Neveu, Yannick Huber, Mattias Frey, Christopher Mayes, Andreas Adelmann (2020)

arXiv

Machine learning assisted non-destructive transverse beam profile imaging

Zhanibek Omarov, Selcuk Haciomeroglu (2021)

arXiv

GAIA: A General AI Assistant for Intelligent Accelerator Operations

Frank Mayet (2024)

arXiv

AI-Enabled Operations at Fermi Complex: Multivariate Time Series Prediction for Outage Prediction and Diagnosis

Milan Jain, Burcu O. Mutlu, Caleb Stam, Jan Strube, Brian A. Schupbach, Jason M. St. John, William A. Pellico (2025)

arXiv

Automated GI tract segmentation using deep learning

Manhar Sharma (2023)

arXiv

Domain Adaptation of Automated Treatment Planning from Computed Tomography to Magnetic Resonance

Aly Khalifa, Jeff Winter, Inmaculada Navarro, Chris McIntosh, Thomas G. Purdie (2022)

arXiv

Superconducting radio-frequency cavity fault classification using machine learning at Jefferson Laboratory

Chris Tennant, Adam Carpenter, Tom Powers, Anna Shabalina Solopova, Lasitha Vidyaratne, Khan Iftekharuddin (2020)

arXiv

Generative Adversarial Networks (GAN) for compact beam source modelling in Monte Carlo simulations

David Sarrut, Nils Krah, Jean-Michel Létang (2019)

arXiv

Machine learning applied to single-shot x-ray diagnostics in an XFEL

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From Compact Plasma Particle Sources to Advanced Accelerators with Modeling at Exascale

Axel Huebl, Remi Lehe, Edoardo Zoni, Olga Shapoval, Ryan T Sandberg, Marco Garten, Arianna Formenti, Revathi Jambunathan, Prabhat Kumar, Kevin Gott, Andrew Myers, Weiqun Zhang, Ann Almgren, Chad E Mitchell, Ji Qiang, David Grote, Alexander Sinn, Severin Diederichs, Maxence Thevenet, Luca Fedeli, Thomas Clark, Neil Zaim, Henri Vincenti, Jean-Luc Vay (2022)

20th Advanced Accelerator Concepts Workshop (AAC'22)

3D deep convolutional neural network segmentation model for precipitate and porosity identification in synchrotron X-ray tomograms

S. Gaudez, M. Ben Haj Slama, A. Kaestner, Manas Vijay Upadhyay (2022)

Journal of Synchrotron Radiation

Leveraging Serial MRI Radiomics and Machine Learning to Predict Risk of Radiation Necrosis in Patients with Brain Metastases Managed with Stereotactic Radiation and Immunotherapy

H. Elhalawani, L.A. Hammoudeh, D.N. Cagney, J.M. Qian, A. Martin, J. Zgrabik, J. Meyers, K.J. Pataki, K. Martin, Y. Khouj, C. Verry, W.L. Bi, O. Arnaout, S.M. Christ, B. Alexander, S. Tanguturi, R. Rahman, D. Haas-Kogan, A.A. Aizer (2022)

64th ASTRO Annual Meeting

Neural networks for rapid phase quantification of cultural heritage X-ray powder diffraction data

Victor Poline, Ravi Raj Purohit, Purushottam Raj Purohit, Pierre Bordet, Nils Blanc, Pauline Martinetto (2024)

Journal of Applied Crystallography

Understanding error propagation in deep learning neural network (DNN) accelerators and applications

Guanpeng Li, Siva Kumar Sastry Hari, Michael B. Sullivan, Timothy Tsai, Karthik Pattabiraman, Joel Emer, Stephen W. Keckler (2017)

Unknown Venue

Survey and Benchmarking of Machine Learning Accelerators

Albert Reuther, Peter Michaleas, Michael Jones, Vijay Gadepally, Siddharth Samsi, Jeremy Kepner (2019)

Unknown Venue

A Survey of Accelerator Architectures for Deep Neural Networks

Yiran Chen, Yuan Xie, Linghao Song, Fan Chen, Tianqi Tang (2020)

Engineering

RBF neural net based classifier for the AIRIX accelerator fault diagnosis

J. C. Ribes, G. Delaunay, J. Delvaux, E. Merle, M. Mouillet (2000)

arXiv

Railway Wheel Impact Force and Alert Prediction Using Machine Learning Models

Gajendra Malviya, Shripad Salsingkar (2025)

AIRO Springer Series

RadiSimCLIP: A Radiology Vision-Language Model Pretrained on Simulated Radiologist Learning Dataset for Zero-Shot Medical Image Understanding

Minhui Tan, Qingxia Wu, Boyang Zhang, Genqiang Ren, Jianlong Nie, Zhong Xue, Xiaohuan Cao, Dinggang Shen (2025)

Lecture Notes in Computer Science

Machine Learning for Precision Dental Diagnosis: Real-Time X-Ray Detection and Patient Record Integration—SmiloScope

M. S. Padmini, Asha Rani Mahadeva, Madhu Nagaraj, Namratha Gopinath (2025)

Information Systems Engineering and Management

A Living Review Pipeline for AI/ML Applications in Accelerator Physics

Adnan Ghribi (2025)

arXiv

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Ghribi, Adnan (2025)

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Preventive Maintenance of Mining Excavators Using Machine Learning: Enhancing Operational Efficiency and Cost-Effectiveness

Syed Adnan Ahmad, Prakash Kumar (2025)

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Steel Surface Defects Recognition Based on Multi-Type Statistical Features and Multiclass Support Vector Machine

Dibya Tripathi, Kheelraj Pandey, A. K. Chauhan (2025)

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Prediction of Tensile Strength and Impact Strength in Fused Deposition Modeling Using a Machine Learning Pipeline

Anne Vogler, Benjamin Küster, Malte Stonis, Ludger Overmeyer (2025)

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TAIDL: Tensor Accelerator ISA Definition Language with Auto-generation of Scalable Test Oracles

Devansh Jain, Marco Frigo, Jai Arora, Akash Pardeshi, Zhihao Wang, Krut Patel, Charith Mendis (2025)

Proceedings of the 2025 58th IEEE/ACM International Symposium on Microarchitecture

BitL: A Hybrid Bit-Serial and Parallel Deep Learning Accelerator for Critical Path Reduction

Seunghyun Lee, Dongho Ha, Sungbin Kim, Sungwoo Kim, Hyunwuk Lee, Won Woo Ro (2025)

Proceedings of the 2025 58th IEEE/ACM International Symposium on Microarchitecture

Bolt Anchorage Defect Identification Based on Ultrasonic Guided Wave and Deep Learning

Hui Xing, Weiguo Di, Xiaoyun Sun, Mingming Wang, Chaobo Li (2025)

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PSVM-MR: A Parallel Support Vector Machine Algorithm Based on MapReduce

Bin-bin Guo, Yimin Mao, A Yaser, Neelakandan Chandrasekaran, Le Kang, Wenhao Li, Decheng Miao (2025)

Communications in Computer and Information Science

DEEP LEARNING-BASED CLASSIFICATION OF SEISMIC DAMAGE ON PARTITION WALL AND ARTIFICIAL CRACKING PATTERN

Kotoka FUJISAWA, Noriyuki TAKAHASHI (2025)

AIJ Journal of Technology and Design

Enhancing RFI management in construction through machine learning-driven predictive models

Neziha Yilmaz, Esin Ergen, Ahmet Muhtar Citipitioglu (2025)

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Digital Twin Framework for PIP-II Linac: AI-Driven Multi-Scale Modeling from Ion Source to 800 MeV

Pathak, Abhishek, Hanlet, Pierrick, Miceli, Tia (2025)

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A “Bottom-Up” Approach for Constructing Underground Engineering Knowledge Graph

Yiming Luo, Yunfei Xiang, Peng Lin, Yong Xia, Yuanguang Liu, Yao Xu, Chaoyi Li (2025)

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Achievement of High■Quality Gallium Oxide Epitaxial Growth via Machine Learning

Yaoping Lu, Yu Zhang, Ben Niu, Titao Li, Zhenjie Zheng, Lemin Jia, Duanyang Chen, Hongji Qi, Kelvin H. L. Zhang, Min Zhu, Haizhong Zhang, Xiaoqiang Lu (2025)

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Unsupervised Online Learning for AC Optimal Power Flow: A Gradient-Guided Physics-Informed Neural Network Approach

Bozhen Jiang, Jing Qu, Qin Wang (2025)

Institute of Electrical and Electronics Engineers (IEEE)

Enhancing Predictive Accuracy in Shear Strength of RC Deep Beams: A Comprehensive Analysis Using Ensemble Machine Learning Models

Arslan Qayyum Khan, Muhammad Huzaifa Naveed, Muhammad Dawood Rasheed, Amorn Pimanmas (2025)

Arabian Journal for Science and Engineering

Two-Step Deep Learning Approach for Fine-Grained Railway Catenary Segmentation

Milo Belien, Metehan Doyran (2025)

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Saffron Adulteration Prediction Based on Fine-Grained Deep Features

Ishrat Nazeer, Ranjeet Kumar Rout, Saiyed Umer (2025)

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Machine-Learning Approach for Pulsed Electromagnetic Field Therapy Parameters Optimization for Enhanced Tissue Penetration

Pooja Kumari Jha, Vikas Kumar, Manoj Kumar Parida, S. Kanagaraj (2025)

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Artificial Intelligence and Machine Learning Based Predictive Modelling of Beam and 3D Bolt Through FEA

Neha M. Deshmukh, Yashwant S. Munde, Surajit S. Wadagaonkar, Avinash S. Shinde, Prashant R. Anerao (2025)

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Zhiar Piroti (2025)

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Anwar Ibrahim, Alexey Petrenko, Maxim Kaledin, Ehab Suleiman, Fedor Ratnikov, Denis Derkach (2025)

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Ibrahim, Anwar, Petrenko, Alexey, Kaledin, Maxim, Suleiman, Ehab, Ratnikov, Fedor, Derkach, Denis (2025)

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Automated beam tuning of the TRIUMF ISAC facility

Shelbaya, Olivier, Jung, Paul, Kester, Oliver, Hassan, Omar (2025)

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Deep Learning-Based Classification of Spine X-Ray Images Using Attention Mechanisms

Asaram Pandurang Janwale, Minal Dutta, Savita Mohurle, Vaduguru Venkata Ramya (2025)

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Abstract 4345868: Machine learning prediction of myocardial ischemia using quantitative assessment of coronary calcifications in non-contrast CT calcium scoring scans

Juhwan Lee, Mohamed H.E. Makhoul, Sadeer Al-Kindi, Ammar Hoori, Tao Hu, Hao Wu, Justin Kim, Sanjay Rajagopalan, David Wilson (2025)

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A data-driven hybrid method combining experiments, finite element modeling and machine learning for impact response prediction of TPU composites

Shunqi Zhang, Luca Lomazzi, Dayou Ma, Andrea Manes (2025)

International Journal of Structural Integrity

Human-in-the-loop reinforcement learning for data quality monitoring in particle physics experiments

Olivia Jullian Parra, Julián García Pardiñas, Lorenzo Del Pianta Pérez, Maximilian Janisch, Suzanne Klaver, Thomas Lehericy, Nicola Serra (2025)

Machine Learning: Science and Technology

ESTIMATIVA DE VIDA ÚTIL REMANESCENTE DE PLACAS CERÂMICAS UTILIZADAS EM FILTRAGEM DE MINÉRIO DE FERRO ATRAVÉS DE MODELO DE CONFIABILIDADE E MÉTODOS DE MACHINE LEARNING APLICADOS A VARIÁVEIS DE PROCESSO INDUSTRIAL DE UM SISTEMA PIMS

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Anais do Encontro Nacional de Engenharia de Produção

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arXiv (Cornell University)

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Ram U. Kalyankar, Kalpak R. Sagar, Vipul M. Patel, Jyotirmay Banerjee, Hemantkumar B. Mehta

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Adwika Singh, Falguni Sharma, Anshuman Shastri (2025)

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Predictive Modeling of ICU Mortality Using Supervised Machine Learning Algorithms

Malithi Upeksha, Sulanie Perera (2025)

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SABER: Symbolic Regression-based Angle of Arrival and Beam Pattern Estimator

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Machine learning-driven optimization of energy absorption in PLA composites reinforced with carbon fibers through FFF additive manufacturing

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Rapid Prototyping Journal

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Gilardi, Antonio, Malyzhenkov, Alexander, Petersson, Alfred, Mostacci Andrea, Pollastro, Andrea, Aksoy, Avni, Filippetto, Daniele, Gamba, Davide, Granados, Eduardo, Tangari, Giacomo, Sjobak, Kyrre, Bonnard, Ladislav, Wroe, Laurence, Carranza-García, Manuel, Franek Ondrej, Korysko, Pierre, Corsini Roberto, Rieker, Vilde, Farabolini, Wilfrid, Mazzoni, Stefano (2025)

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Advancing accelerator science through data-intensive research and training

Welsch, Carsten (2025)

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Machine learning approach to MDI optimization for 3 TeV c.o.m. Muon Collider

Castelli, Luca (2025)

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Advances in machine learning inference of dynamic aperture evaluation for the LHC

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Sun Yue, Bellandi Andrea, Eichler, Annika, Richter, Bozo, Diomedes, Marco, Schmidt, Christian, Schlarb Holger, Branlard, Julien, Herrmann, Max (2025)

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Haymanot Derebe Bizuneh (2025)

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Ujunwa Mgbob, Rafi Sultan, Dongxiao Zhu, Joshua Kim (2025)

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Papoyan, V., Aparin, A., Ayriyan, A., Grigorian, H., Korobitsin, A. (2025)

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ChatEED: An agentic retrieval assistant for accelerator operators

Aaron Zachary Reed, Claudio Bisegni, Sandesh Shrestha, Michelle Huang, Daniel Ratner (2025)

Proceedings of the SC '25 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis

Machine Learning in Semiconductor Failure Analysis: Techniques and Case Studies (2025 Update)

Michael Kögel, Sebastian Brand, Frank Altmann (2025)

International Symposium for Testing and Failure Analysis

NNia-8: An 8-Core RISC-V Neural Network Inference Accelerator with Efficient Processing Elements and Memory Utilization

Xingbo Wang, Yucong Huang, Xinyu Kang, Yuru Li, Qi Wang, Terry Tao Ye (2025)

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R. Anand, D. P. Monish, D. Madhukar, C. G. Paari (2025)

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Deep Learning Assisted Modeling for χ^2 Nonlinear Optics

Jack Hirschman, Erfan Abedi, Minyang Wang, Hao Zhang, Abhimanyu Borthakur, Justin Baker, Andrea L. Bertozzi, Randy Lemons, Sergio Carbajo (2025)

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Chao Guo, Youhua Shi (2025)

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Machine learning with ultrasound examination for prediction of intraoperative hypotension during robot-assisted radical prostatectomy

V. S. Andreenkov, A. V. Vlasenko, A. N. Kornienko, A. S. Kazakov, E. P. Rodionov, K. B.

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Medical alphabet

Categories of Defect Prediction

Zhou Xu (2025)

Machine-Learning-Assisted Software Defect Prediction

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Parameter estimation of three-phase voltage source converter using physics-informed machine learning

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Measurement Science and Technology

Automated RF Phase Adjustment for Beam Stabilization in the Fermilab Linac

R. R. Chichili, J. A. Sulskis, R. Sharankova, B. Vamanan, S. Ravi (2025)
arXiv

Fast Adaptive Neural Control of Resonant Extraction at Fermilab

A. Whitbeck, J. Berlioz, K. Danison-Fieldhouse, K. Hazelwood, M. Khan, J. Mitrevski, A. Narayanan, J. St. John, N. Tran, J. Ji, M. Walter (2025)
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Hybrid deep semantic query expansion using multi-objective fireworks-transformer optimization: a reinforcement learning approach

Fouad Bekkari, Med Redouane Kafi, Amine Khaldi (2025)
Evolutionary Intelligence

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Chichili, R.R., Sulskis, J.A., Sharankova, R., Vamanan, B., Ravi, S. (2025)
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Study on estimation of internal cracks using electromagnetic waves and machine learning

Hinata Yamamoto, Hitoshi Ito, Toshiaki Mizobuchi (2025)
Proceedings of the Institution of Civil Engineers - Construction Materials

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Papun Mondal, Sanjoy Kumar Pal, Pradipta Panchadhyayee, Suman Kumar Dash, Subhash Chandra Samanta (2025)
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Anthropologischer Anzeiger

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International Journal of Particle Therapy

First Experimental Demonstration of Machine Learning-Based Tuning on the PSI Injector 2 Cyclotron

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Timothy Nunan (2025)

International Journal of Middle East Studies

Photonic Quantum-Accelerated Machine Learning

Rambach, Markus, Roy Abhishek, Gilchrist, Alexei, Sakurai, Akitada, Munro, William J., Nemoto Kae, White, Andrew G. (2025)

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Autonomous Pressure Control in MuVacAS via Deep Reinforcement Learning and Deep Learning Surrogate Models

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Machine learning-based prediction of magnet errors in storage ring light sources

Jianhao Xu (2025)

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